# In-Vehicle AR-HUD System to Provide Driving-Safety Information

Hye Sun Park, Min Woo Park, Kwang Hee Won, Kyong-Ho Kim, and Soon Ki Jung

**Augmented reality (AR) is currently being applied actively to commercial products, and various types of intelligent AR systems combining both the Global Positioning System and computer-vision technologies are being developed and commercialized. This paper suggests an in-vehicle head-up display (HUD) system that is combined with AR technology. The proposed system recognizes driving-safety information and offers it to the driver. Unlike existing HUD systems, the system displays information registered to the driver's view and is developed for the robust recognition of obstacles under bad weather conditions. The system is composed of four modules: a ground obstacle detection module, an object decision module, an object recognition module, and a display module. The recognition ratio of the driving-safety information obtained by the proposed AR-HUD system is about 73%, and the system has a recognition speed of about 15 fps for both vehicles and pedestrians.**

**Keywords: AR, augmented reality, HUD, head-up display, vehicle and pedestrian recognition, stereo camera, SVM, support vector machine, HOG, histograms of oriented gradients, driving-safety information, in-vehicle intelligent information system.**

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## I. Introduction

Augmented reality (AR) is a computer graphics technology that supplements a real environment with virtual objects or information based on computer technology. AR is currently being applied in many aspects of the real world [1]-[8]. In the Republic of Korea in particular, as of 2011, the number of smartphone users has surpassed twenty million people. Accordingly, companies focusing on AR technology are actively progressing. According to Juniper Research [9], the AR market has increasingly expanded, and it is expected that the technology will earn more than seventy billion US dollars by 2014. This technology is therefore expected to be applied to fields beyond the area of smartphones.

In 1999, GM started the development of a vehicle-installed head-up display (HUD). Thereafter, other famous car companies, such as BMW and Peugeot, started to develop HUD-based vehicles. However, most of the information offered was already available on the dashboard, such as the driving speed and vehicle direction. AR technology was first used in vehicles in 2010 due to the rapid development of the smartphone but only provided limited information to a designated area on a vehicle's windshield [10], [11].

However, thanks to the intellectualization and radical development of vehicle technology, technologies that can provide the driver with a large amount of information within the vehicle have been developed. The driver can therefore receive not only information on their own vehicle but also information on the environment in which they are driving and information on their physical state and mental state, such as driving habits, inattention (drowsiness or gaze distribution), and even health status.

Past research has indicated that the vast majority of motor




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vehicle crashes are caused by human error. According to a study released by the National Highway Traffic Safety Administration (NHTSA) of the US Department of Transportation and the Virginia Tech Transportation Institute (VTTI), 80% of crashes in the United States involve some form of driver distraction. Thus, an alarm that alerts the driver of an obstacle in the road helps to reduce vehicle accidents. Such an alarm is especially important to prevent collisions with pedestrians or other cars. In this regard, many such technologies and products are being developed around the world, and driving-safety information is now being offered to the driver through various modalities. Thus, driving workloads, including visual, auditory, psychomotor, and cognitive, arise when the driver recognizes information transferred through touch sensors, in-vehicle terminals, and so on [12]-[18]. In particular, the driver receives 80% of this information visually while looking forward at the road to support safe driving practices [19], [20]. If augmented driving-safety information matched with the information regarding the real environment is provided while the driver's gaze is toward the driving direction, the driver can recognize the information naturally and focus on driving safely.

Thus, this paper introduces an in-vehicle AR-HUD system that detects such driving-safety information and offers the information to the driver, fitting the driver's viewpoint, as shown in Fig. 1. In particular, the proposed system intelligently judges what information is proper to provide based on road attributes associated with vehicle traffic and those associated with pedestrian traffic, using the number of lanes, road type, and vehicle speed. To improve the performance speed in real time, the proposed system operates in one of two modes when both modes are not needed, such as on a vehicle-only road.

In-vehicle AR-HUD technologies are being actively developed [21]-[24]. In particular, prominent automobile companies are selling high-quality vehicles installed with AR-HUD products. However, the current commercially available AR-HUD products all offer information in front of the driver but in a state that does not match the real environment. In addition, simple vehicle information (speed, RPM, and so on) and route information is displayed in a much more visually distracting way than in a head-down display.

If such driving-safety information as pedestrian and vehicle information is offered in front of the driver in a state that does not match the real environment, it can interrupt the driver's concentration, causing confusion. Therefore, it is best that the interface through which driving-safety information is provided appears within the driver's forward-facing view in a state that matches the real environment. In this respect, since 2010, GM has been conducting research to provide driving-safety information for traffic signs and lanes through an interface that superimposes the data onto the real environment in a position that accommodates the driver's view [25].

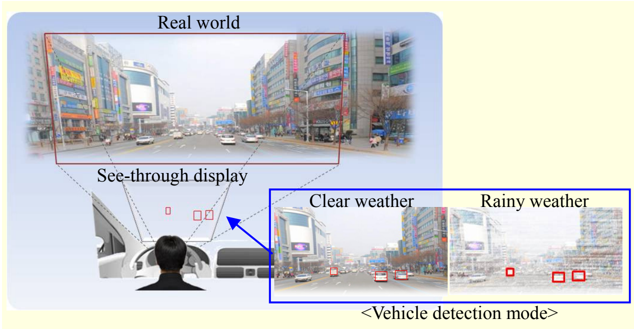


Fig. 1. Concept map of proposed system.

As mentioned above, the proposed method detects the most important vehicle and pedestrian information and provides it to the driver. Many technologies for detecting such information while driving have been previously developed [26]-[32].

In the proposed system, stereo cameras are used to extract more correct distance values and reduce the recognition times. An obstacle on the ground plane is extracted from the visual data from the stereo cameras, and other vehicles and pedestrians are distinguished from the detected obstacle. Then, since the proposed system finds obstacles from the input images using ground detection and the system thereafter extracts the candidate region and finds the driving-safety information only in the extracted region, it can reduce the detection area required as compared to existing methods based on a single camera that detects all areas of the input images. The proposed system is therefore developed to be easily combined with real vehicles by reducing the recognition time.

## II. In-Vehicle AR-HUD System for Providing Driving-Safety Information

### 1. Necessity for Development of AR-HUD for Driving-Safety

To reduce driving accidents, driving-safety needs to be modeled. Among the various models, one suggested in Australia consists of four factors: safe speed limit, vehicle safety, road-facility safety, and driver safety [33].

The first factor, a safe speed limit, applies to driving safety in that the driver is expected to maintain a safe distance behind the car in front by maintaining a proper vehicle speed. The next factor, vehicle safety, refers to the utilization of mounted devices used to promote accident prevention. The third factor refers to the establishment of safety facilities to reduce accidents at known accident sites or crossways. Finally, the fourth factor refers to ensuring the protection of the driver through driver education.

To maintain a safe distance behind other vehicles and avoid collisions, the proposed system informs and warns the driver of detected obstacles using various mounted devices, under consideration of the previously mentioned factors. Through this system function, the number of traffic accidents can be significantly reduced.

However, most existing devices provide the driver with warning sounds using auditory sensors. In recent years, such devices have offered information through vibrations using the development of a tactile sensor. However, more than 80% of the information is received visually by the driver while driving. In addition, for driver safety, the driver must keep their gaze forward as much as possible. The literature has shown through simple response experiments that reaction times are lowest (fastest) for auditory modality (approximately 150 ms), followed by touch (approximately 155 ms), with visual modality being the slowest (approximately 190 ms). However, the transmission of information through vibrations has a limitation during driving and may actually interfere with the driver. Thus, tactile warning devices can be used only under special and emergency situations according to the context of the information, such as during driver drowsiness. The driver may sometimes ignore the provided information from auditory devices when concentrating on driving.

Previously proposed systems are not equipped to accurately recognize the location of a detected obstacle. Regarding obstacles that pose a high threat, it is crucial that location information be available in the driver's field of view to effectively avoid such obstacles.

## 2. Overview of Proposed System

The hardware and software of the proposed system are composed as shown in Fig. 2. Two stereo cameras are mounted on the rearview mirror and receive the frontal images in real time, with a viewing angle of 65 degrees. PC software installed in the vehicle detects other vehicles and pedestrians from the input images, and the detection results are provided to the driver through a see-through display installed in the windshield, fitting the driver's line of vision. The proposed system detects all vehicles and pedestrians in front of the vehicle and provides the results to the driver.

However, future intelligent AR-HUD systems will filter the detected information and offer only the information useful to the driver according to the provision type (modality) and information representation (display) method for easier understanding. Based on the problems mentioned above, current AR-HUD systems should be further developed, considering driver safety. For the purpose of convenience in this experiment, the proposed system is designed to use a see-through display installed in front of the passenger seat, fitting the line of vision of the person in the passenger seat. If the proposed system is developed to completion for the commercial market, the display will instead be installed on the driver's side.

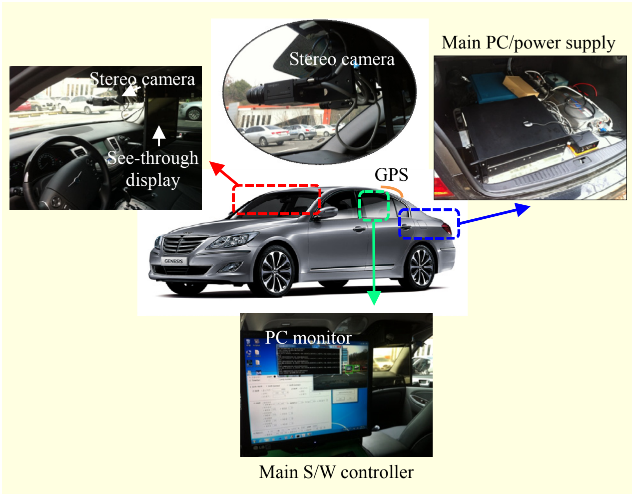


Fig. 2. Overview of proposed system.

## 3. System Configuration

Figure 3 shows the configuration of the proposed system. The system consists of four modules. Each module is described below along with each submodule.

### *A. Ground Obstacle Detection Module*

All objects on a road can be potential obstacles. In this system, we first extract all obstacles using their geometric properties. We also classify them into "significant" and "less significant" objects based on the mode selection module, which is triggered under certain circumstances. Significant objects are identified using specialized detectors (that is, vehicle and pedestrian detectors).

We use a billboard sweep stereo algorithm to identify obstacles on the ground plane [34]. Unlike many other algorithms that estimate the ground plane after disparity matching, the algorithm determines the ground, obstacles, and background regions, as well as their corresponding slopes and distances, with reliable accuracy. Among the three categories stated above, we use obstacle regions and their disparity values for further processing. To achieve a near real-time performance, we implement the algorithm on a graphics processing unit (GPU) and use input at half resolution.

The billboard sweep stereo algorithm consists of two main steps: matching the cost computation and optimization. It


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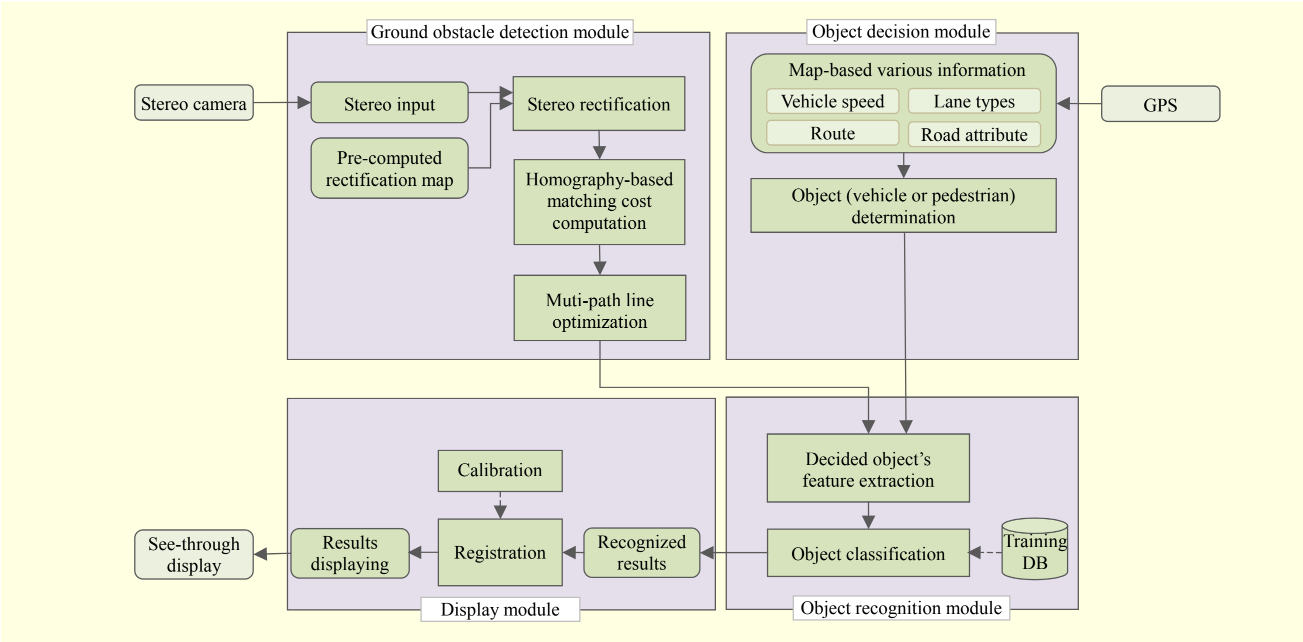


Fig. 3. Configuration of proposed system.

utilizes homography-based matching costs using pre-calibrated candidate planes, which represent varying slopes of the ground plane and distances of the billboard planes. For the next step, the algorithm performs multipath line optimization over the cost volume to select the disparity values and slopes as well as the labels. Billboard labels are considered detected obstacles, and the corresponding disparity values are used as the distance to the detected objects.

Therefore, the proposed billboard approach is a road-based vehicle detection system. It firstly detects the road geometry and computes the depth of obstacles, including the height from the road plane. In this case, we assume that all obstacles are dangerous. Therefore, we reduce the false positive for vehicle detection.

### B. Object Decision Module

This module selects the detection object using the surrounding environment, such as the vehicle speed or road type in real time. If the system does not need to find other vehicles and pedestrians at the same time, only the selected mode is used. This therefore reduces the computation cost. This module, based on driving-situation cognition, selects and provides the appropriate information. Thus, this module is a dynamic and intelligent module for providing driving-safety information.

### C. Object (Vehicle/Pedestrian) Recognition Module

If a detected object between the vehicle and a pedestrian is determined through context awareness in the previous module, the proposed system should detect the object. For this module, the input images and ground obstacle images are needed for detection. A ground obstacle is computed using the billboard sweep stereo matching algorithm by the ground obstacle detection module [35]. Obstacles are deemed dangerous according to height. Therefore, we can recognize the partial obstacles using the billboard sweep stereo matching algorithm even if the detection fails. The vehicle and pedestrian recognition module consists of three steps: 1) candidate region detection, 2) GPU-based histograms of oriented gradients (HOG) extraction, and 3) support vector machine (SVM) training and classification.

First, we generate the hypothesis using the edge-based segmentation algorithm. In the edge-based segmentation algorithm, we first perform canny edge detection and find the horizontal and vertical lines through a variance of intensity from the edge image using a generalized projection function [36]. After we make the cell region using the detected lines, we extract the candidate regions based on the average disparity for each segmented cell. In this step, the extracted region is a candidate location for a vehicle or pedestrian.

Next, we extract the HOG features for the detected candidate region. An HOG feature is used as the main feature to detect other vehicles and pedestrians. In this step, we use GPU acceleration to extract the feature descriptor in real time. In general, HOG feature extraction consumes a lot of execution time in its computation. Therefore, we use a GPU to improve the speed of HOG feature extraction for vehicle and pedestrian

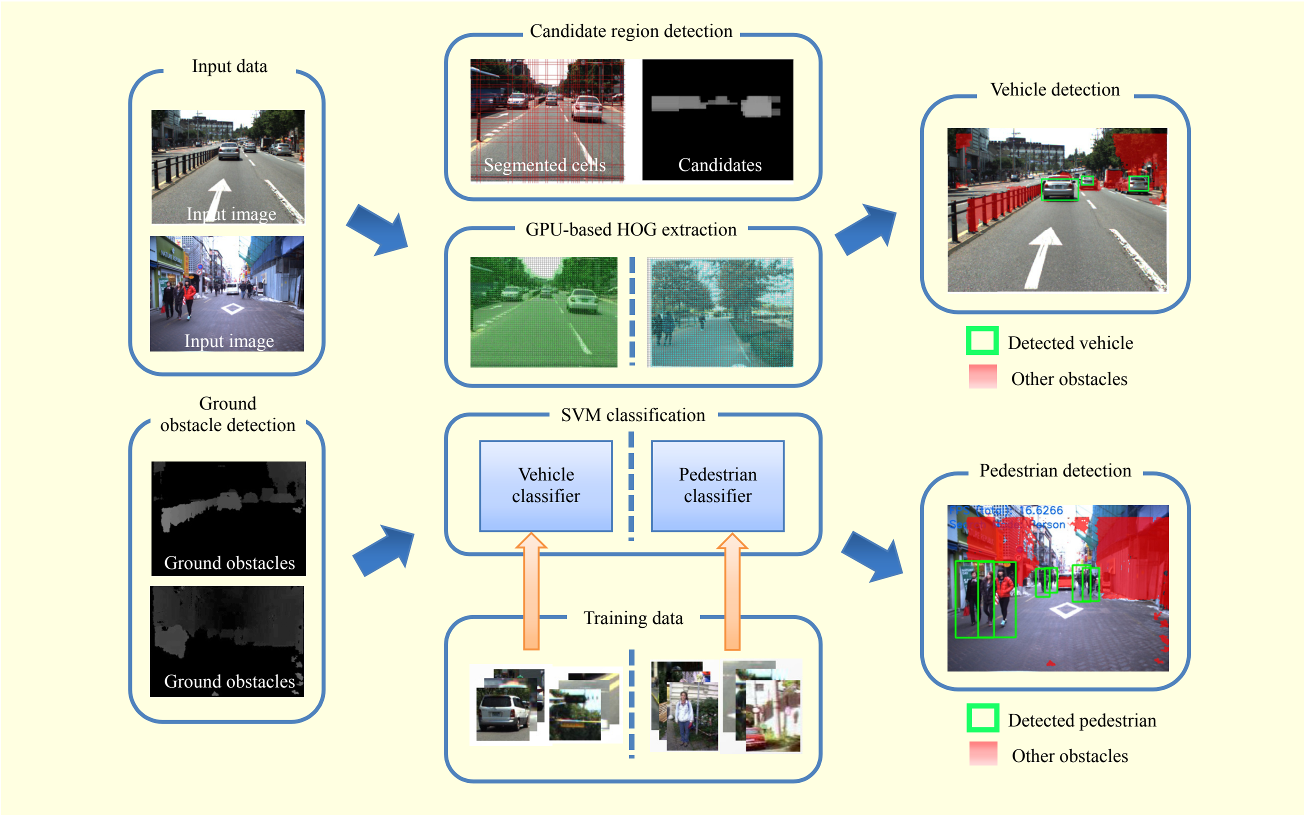


Fig. 4. Object recognition module.

detection.

Finally, we perform a hypothesis verification using two types of linear SVM classifiers [37], [38]. The two types of linear SVM classifiers are chosen based on the selected mode of the previous module. The proposed system performs recognition for the detected candidate region using the selected classifier. If the recognition is complete, we determine and display the vehicle or pedestrian region. Each linear SVM classifier is learned using training data composed of positive and negative data before the system is performed.

Figure 4 shows the process of the vehicle and pedestrian recognition module. In Fig. 4, we illustrate the detected regions and non-detected obstacle regions.

The proposed method uses the two steps of hypothesis generation and hypothesis verification. The method cannot increase the processing time. First, we perform the depth grouping using a connected component labeling algorithm to reduce the computational region in the hypothesis generation step. Next, we perform the fast classification using a simple linear SVM method in hypothesis verification. Furthermore, we perform the tracking and cannot perform the hypothesis verification for detected regions by tracking in the next frame. Therefore, the proposed method is not affected even if the number of candidates is increased.

*D. Display Module*

The system receives the location values ($x$, $y$ coordinates) of pedestrians or vehicles and their distance values, which were detected from the previous modules through a TCP/IP socket. It also shows the vehicle and pedestrian information detected from the input values in the form of a square box, fitting the view of the driver.

To make the information correctly match the real environment according to the variations in the driver's view, this system needs to use a rigorous approach [39] to present the pedestrian or vehicle information in a 3D space onto an HUD display through 3D rendering. For this, the position and eye location of the driver should be calculated exactly.

However, while this method is accurate, the calculations are complex. Therefore, the system adopts an approximation method [39] with minimum complexity and can match and present the information in real time. The method only conducts geometric modeling between the 2D image of a camera and a 2D HUD machine on the assumption that the position and view of the driver are fixed toward the front. Since this system converts a 2D image into an HUD plane, it is therefore carried out in a projective transformation (2D homography) calculated by (1):

$$x_h = \frac{h_{11}x_c + h_{12}y_c + h_{13}}{h_{31}x_c + h_{32}y_c + h_{33}},$$
$$y_h = \frac{h_{21}x_c + h_{22}y_c + h_{23}}{h_{31}x_c + h_{32}y_c + h_{33}}, \tag{1}$$

where, $x_c$, $y_c$ are the image coordinates in the camera and $x_h$, $y_h$ are the image coordinates in the HUD. Additionally, $h_{11}$ through $h_{33}$ are projective transformation parameters, each of which has eight degrees of freedom. To calculate the projective transformation parameter, it is necessary to specify more than four point correspondences. In general, an automatic matching method, such as SIFT or SURF, is used to correspond feature point pairs. However, we manually extract the corresponding points between the camera image and the HUD image in the proposed system. To determine the projective transformation parameter given a set of 2D to 2D point correspondences between camera and HUD, we use the direct linear transformation (DLT) algorithm; then, we assume the parameters follow to affine transformation. Thus, the DLT algorithm is easily adapted to enforce the condition by setting $h_{31}$=$h_{32}$=0 and $h_{33}$=1. To estimate the remaining $h$ values, (1) is translated into (2), and (3) is then extracted using a least squares method.

$$\begin{bmatrix} x_c & y_c & 1 & 0 & 0 & 0 & -x_h x_c & -y_c x_h \\ 0 & 0 & 0 & x_c & y_c & 1 & -y_h x_c & -y_h y_c \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} + \boldsymbol{r} = \begin{bmatrix} x_h \\ y_h \end{bmatrix}, \tag{2}$$

where $H \in \{h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}\}$, a coefficient matrix of $H$ is defined by $A$, and $\boldsymbol{r}$ is a residual vector.

$$H = \left(A^T \quad A\right)^{-1}\left(A^T \quad \begin{bmatrix} x_h \\ y_h \end{bmatrix}\right). \tag{3}$$

# III. Experiments

## 1. Experimental Environments

For vehicle recognition, we perform an experiment in which a vehicle equipped with the proposed system drives at a speed of 70 km/h to 80 km/h in more than two lanes, and detects other vehicles within 40 m to 45 m of its radius. For the pedestrian recognition, we perform an experiment in which a vehicle equipped with the proposed system drives at a speed of less than 30 km/h along a narrow road in a residential district and detects pedestrians within a range of around 30 m of its radius. The cameras used for this experiment are GS2-FW-14S5 models from the Point Grey Research Company, which are 8 mm cameras with a resolution of 1038 × 1036 and can obtain an image at a speed of 30 fps. In addition, we use IEEE 1394b for the interface.

(a)

(b)

Fig. 5. Example of display used in proposed detection results of (a) other vehicles and (b) pedestrians.

We use a 22-inch transparent Samsung LCD display with a transparency of 15%. Figure 5 shows the images of the vehicles and pedestrians detected through the proposed system on the display.

## 2. Experiment Results

The recognition ratio [40] and recognition speed [41] of this system are calculated using (4) and (5).

$$\text{Recognition rate}(\%) = \frac{\text{true positive}}{(\text{true positive+false positive})}, \tag{4}$$

$$\text{fps} = \frac{\text{tickfrequency}}{(\text{end tickcount-start tickcount})}. \tag{5}$$

The results of the vehicle recognition are shown in Table 1. The experiment is conducted in various experimental environments, as shown in Fig. 6.

During vehicle recognition, we comparatively analyze the system using two road types: urban road and highway. An

Table 1. Vehicle recognition results.

| Vehicle | Urban road | Highway | Highway in rain |
|---|---|---|---|
| Recognition rate (%) | 72.86 | 82.17 | 61.50 |
| Performance (fps) | 13.97 | 15.21 | 15.60 |

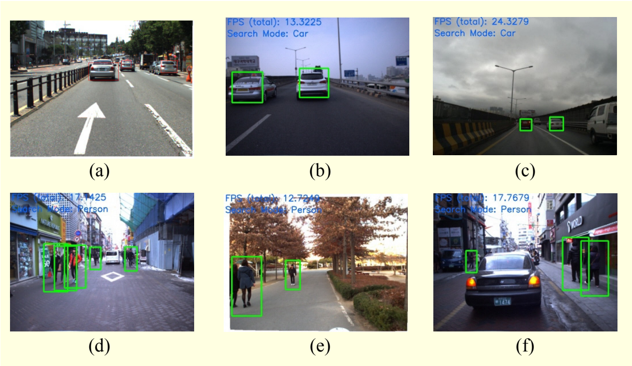


Fig. 6. Vehicle recognition results for (a) urban road, (b) highway, and (c) highway in rain. Pedestrian recognition results for (d) downtown street, (e) residential street, and (f) urban road.

urban road is a general road with cars and people walking on a sidewalk or crossing through a crosswalk. A highway is usually a vehicle-only road. In the case of the highway, we conduct a comparative analysis according to the weather conditions. The proposed method performs the vision-based forward obstacle detection. It performs the alert and notifies the warning for collision information. The vision-based method cannot generally detect the obstacles in severe weather. However, we perform the experiments to measure the robustness of the proposed method in rainy weather. Table 1 shows the results of the vehicle recognition experiments in sunny and rainy weather. The system is more robust on the highway than on the urban road. The reason for this is that the surrounding environment is more complex on an urban road. In addition, because the wipers are automatically turned on when raindrops hit the windshield, the recognition rate is lower than on sunny days.

In pedestrian recognition, we comparatively analyze the system using three road types: downtown street, residential street, and urban road. A downtown street is a city street with cars and people. On such roads, people walk together in groups, and the recognition rate is therefore slightly lower. A residential street indicates a pedestrian-oriented road or a road with a school or similar type of building. On residential streets, pedestrians usually do not walk together, and the highest recognition rates are therefore achieved in comparison with other roads. Finally, an urban road is the same type of road as the above-mentioned urban road used for vehicle recognition.

Table 2. Pedestrian recognition results.

| Pedestrian | Downtown street | Residential street | Urban road |
|---|---|---|---|
| Recognition rate (%) | 73.48 | 81.80 | 68.21 |
| Performance (fps) | 14.03 | 16.71 | 14.60 |

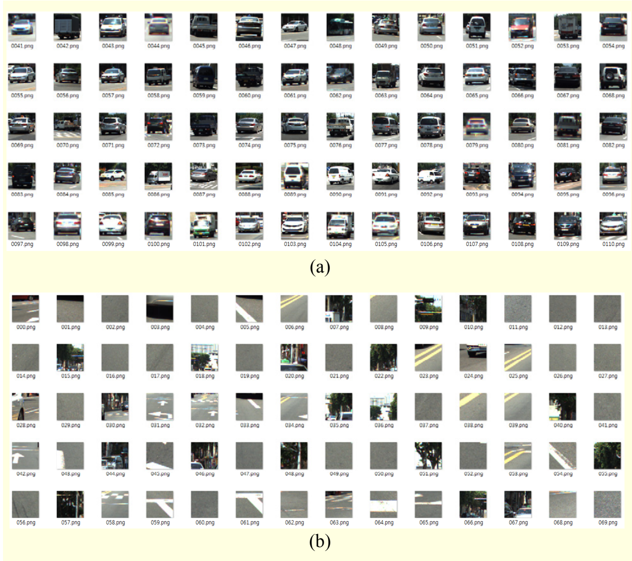


Fig. 7. Training dataset for recognition in proposed system: (a) positive and (b) negative data for training vehicle data.

In such roads, the driver's view of the pedestrian may be obscured by the vehicle in front of the driver because pedestrians mainly exist along the side of the road. Therefore, the recognition rate for such roads is the lowest. Table 2 shows the results of the pedestrian recognition experiments. The system is more robust on a residential street than on the other types of road because the background environment is simpler and it is less likely that people walk in groups.

Figure 7 shows the images used in the learning data for the vehicle recognition module. A total of 217 images are used as positive data, and 410 images are used as negative data. The learning data used in the pedestrian recognition module is from the INRIA person dataset, with 2,416 images used as positive data and 12,180 images used as negative data.

## IV. Conclusion

The proposed system, in which an HUD system is combined in a vehicle with AR technology, matches augmented driving-safety information with the information on other vehicles and pedestrians in the real world and offers the information to the driver in real time through a transparent display installed in front of the driver, fitting the driver's view.

The proposed method seeks out the ground from the images input from stereo cameras and detects the foreground region by removing the ground from the image. Within the detected foreground region, the system then detects pedestrians and vehicles based on their feature points, learns and classifies the detected objects through an SVM, and recognizes the vehicles and pedestrians from the input images in real time. In this paper, we proposed an obstacle detection method based on the road geometry. The method reduces the false positive better than other detection methods do because we can detect partial obstacles using the billboard sweep stereo algorithm. Furthermore, we use a GPU acceleration to improve the performance of the proposed system. When we tested the system on actual roads, we achieved a 72% recognition rate regarding other vehicles and a 74% recognition rate regarding pedestrians. To increase the recognition rate of the proposed system, we need to collect more data by changing the illumination, vehicle speeds, and other environmental conditions.

The proposed system can perform real-time obstacle detection in sunny or rainy weather. However, it has not been verified that the proposed system can likewise perform in severe weather or in the nighttime. Therefore, we will extend the obstacle detection to work in invisible environments through a fusion of night vision technology and radar sensors.

In addition, this proposed system detects all pedestrians and vehicles from the input images within a certain camera range. Sometimes, driver distraction is caused by oversaturation of information. Therefore, a future system is needed to filter the input information and obtain only the information that can help protect the driver from accidents. For example, a pedestrian who crosses the road or makes a sudden movement toward the vehicle would rate high on a scale of danger, whereas a pedestrian simply walking in the same direction as the moving vehicle would rate low on a scale of danger. Thus, a system developed in the future must be able to anticipate such dangers based on a model that can predict the movements of pedestrians and offer such driving-safety information to the driver adaptively, according to the level of risk.

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